

1a)

Original array:

83 86 93 86 92 21 90 59 26 26 11 68 67 77 15 35 49 62 27 63 40 72 36 29 82

Sorted array (may be incorrect due to data hazard):

11 15 21 26 26 27 29 35 36 40 49 59 62 63 67 68 72 77 82 83 86 86 90 92 93

1b)

Original array:

83 77 93 86 62 59 26 40 36 68 67 29 82 86 15 35 92 49 21 27 90 63 26 72 11

Sorted array (data hazard eliminated):

11 15 21 26 26 27 29 35 36 40 49 59 62 63 67 68 72 77 82 83 86 86 90 92 93

2)

Element 42 found at index 42.

3)

Fibonacci sequence (sequential):

0 1 1 2 3 5 8 13 21 34 55 89 144 233 377 610 987 1597 2584 4181

Fibonacci sequence (parallel):

0 1 1 2 3 5 8 13 21 34 55 89 144 233 377 610 987 1597 2584 4181

4)

No free functional unit available for instruction 3

No free functional unit available for instruction 4

No free functional unit available for instruction 6

No free functional unit available for instruction 1

Executed instruction 2: Result = 20

Executed instruction 5: Result = 9

5)

Thread 1 starting...

Thread 1 doing section 1

Thread 1: c[0]= 22.350000

Thread 1: c[1]= 24.850000

Thread 1: c[2]= 27.350000

Thread 1: c[3]= 29.850000

Thread 1: c[4]= 32.349998

Thread 1: c[5]= 34.849998

Thread 1: c[6]= 37.349998

Thread 1: c[7]= 39.849998

Thread 1: c[8]= 42.349998

Thread 1: c[9]= 44.849998

Thread 1 doing section 2

Thread 1: d[0]= -22.350000

Thread 1: d[1]= -21.850000

Thread 1: d[2]= -21.350000

Thread 1: d[3]= -20.850000

Thread 1: d[4]= -20.350000

Thread 1: d[5]= -19.850000

Thread 1: d[6]= -19.350000

Thread 1: d[7]= -18.850000

Thread 1: d[8]= -18.350000

Thread 1: d[9]= -17.850000

Number of threads = 2

Thread 0 starting...

Thread 0 done.

Thread 1 done.

6)

Sum = 30.000000

7)

Starting matrix multiple example with 2 threads

Initializing matrices...

Thread 0 starting matrix multiply...

```

Thread=0 did row=0
Thread=0 did row=1
Thread=0 did row=2
Thread=0 did row=3
Thread=0 did row=4
Thread 1 starting matrix multiply...
Thread=1 did row=5
Thread=1 did row=6
Thread=1 did row=7
Thread=1 did row=8
Thread=1 did row=9

```

Result Matrix:

```

0.00 140.00 280.00 420.00 560.00 700.00 840.00
0.00 168.00 336.00 504.00 672.00 840.00 1008.00
0.00 196.00 392.00 588.00 784.00 980.00 1176.00
0.00 224.00 448.00 672.00 896.00 1120.00 1344.00
0.00 252.00 504.00 756.00 1008.00 1260.00 1512.00
0.00 280.00 560.00 840.00 1120.00 1400.00 1680.00
0.00 308.00 616.00 924.00 1232.00 1540.00 1848.00
0.00 336.00 672.00 1008.00 1344.00 1680.00 2016.00
0.00 364.00 728.00 1092.00 1456.00 1820.00 2184.00
0.00 392.00 784.00 1176.00 1568.00 1960.00 2352.00

```

Done.

8)

1st parallel region:

thread 0: a,b,x = 0 0 1.000000

thread 1: a,b,x = 1 1 2.100000

master thread doing serial work here

2nd parallel region:

thread 1: a,b,x = 1 0 2.100000

thread 0: a,b,x = 0 0 1.000000

9)

2 threads

factorial(0) = 1 x[0] = 1

factorial(1) = 1 x[1] = 1

factorial(2) = 2 x[2] = 3

factorial(3) = 6 x[3] = 18

factorial(4) = 24 x[4] = 180

factorial(5) = 120 x[5] = 2700

factorial(6) = 720 x[6] = 56700

factorial(7) = 5040 x[7] = 396900

factorial(8) = 40320 x[8] = 5953500

factorial(9) = 362880 x[9] = 142884000

factorial(10) = 3628800 x[10] = 563088704

factorial(11) = 39916800 x[11] = -430812096

user43@csdept-HP-dx2480-MT-KL969AV:~\$./a.out 4

4 threads

factorial(0) = 1 x[0] = 1

factorial(1) = 1 x[1] = 1

factorial(2) = 2 x[2] = 3

factorial(3) = 6 x[3] = 18

factorial(4) = 24 x[4] = 24

factorial(5) = 120 x[5] = 216

factorial(6) = 720 x[6] = 3240

factorial(7) = 5040 x[7] = 22680

factorial(8) = 40320 x[8] = 340200

factorial(9) = 362880 x[9] = 8164800

factorial(10) = 3628800 x[10] = 81648000

```
factorial(11) = 39916800    x[11] = 1714608000
```

```
10)
```

```
2 threads max
```

```
6.83 seconds
```

```
5761455 primes between 0 and 100000000
```